

Digitally Calibrated 768-kS/s 10-b Minimum-Size SAR ADC Array With Dithering

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Abstract—Array sensors require a high-performance analog-to-digital converter (ADC) array with small area and low power. Successive-approximation register (SAR) ADC has good potential for ADC array due to its simple analog circuits. However, SAR ADCs with 10-b resolution and higher normally need a large capacitor array due to the stringent matching requirement. The large capacitor array also limits the ADC dynamic performance. The capacitor mismatch has been compensated by analog calibration techniques. In this work, a novel digital calibration method is developed for SAR ADC based on dithering. With dithering, weights of most significant bit (MSB) capacitors can be measured accurately so that very small capacitors can be used in the SAR ADC due to the relaxed matching requirement. A modified bit-cycling procedure is developed to avoid the code gaps caused by capacitor dithering. This calibration technique requires no analog calibration overhead and simple digital decoders. The technique is implemented in an ADC array design including 256 SAR ADCs for a high-speed CMOS imaging sensor in a 0.18- μm CMOS process. The 10-b SAR ADC is designed with the minimum capacitor array size in the process. A single SAR ADC only occupies $15\ \mu\text{m} \times 710\ \mu\text{m}$. Sampling at 768 kS/s, peak DNL and peak INL of the original ADCs averaged across the array are 0.82 least significant bit (LSB) and 3.85 LSB, respectively. For a signal close to the Nyquist frequency, original ADCs have 7.96-b average ENOB. After calibration with dithering, ADCs have 0.55-LSB peak DNL and 0.77-LSB peak INL averaged across the array. The average ENOB improves to 9.83 b. Compared with the benchmark 10-b SAR ADCs, this design is the most area-efficient design. In this work, the calibration decoders are implemented off-chip. With a sample-and-hold amplifier, the calibration method can run in the background.

Index Terms—ADC array, analog-to-digital converter (ADC), digital calibration, dithering, successive-approximation register (SAR).

I. INTRODUCTION

ARRAY sensors are widely applied in various applications, such as imaging [1], bio-potential acquisition [2] and DNA sensing [3]. Analog signals from a large-scale array sensor can be quantized by one high-speed Nyquist-rate analog-to-digital converter (ADC), such as pipelined ADC.

However, with increasing of the pixel counts and the frame rate of the array sensor, such an approach becomes less feasible. High-speed ADCs are power hungry with sampling rate of several hundreds of MS/s [4]. Moreover, the high-speed clock could couple to the analog sensor circuits [5]. Hence, more frequently, array sensors are organized in columns and quantized by ADC arrays ([4], [6]–[11]).

As hundreds of channels are normally included in the array sensor, the size, reliability and power consumption of individual ADC are important for the ADC array. Single-slope ADCs are small in size. They are commonly used in slow array sensors with low-resolution data ([4], [6], [7]). However, slope ADCs have high over-sampling clock ratio. For sensors sampled at 100 kS/s with 10-b resolution or higher, successive-approximation register ADCs (SAR ADCs) [8], [9] and cyclic ADCs [10], [11] are normally designed. Similar to pipelined ADCs, cyclic ADCs require high-gain and high-speed operational amplifiers (OPAMPs) for good linearity and short settling time ([12], [13]). Hence, high power and high analog circuit complexity are needed. With the scaling of the fabrication technology, it is challenging to achieve high-gain OPAMPs [14]. Also, in a cyclic ADC array, mixed-signal circuits are confined within a small column area, which is not favorable for the performance of analog circuits [15] as the ADC array area normally has serious power supply injections.

SAR ADCs have been used for column quantization in high-speed CMOS imaging sensors (CIS) [8], [9]. SAR ADC requires mainly digital circuits and simple analog circuits. It has been actively researched in recent years to achieve excellent power efficiency [14], [16]–[25]. Nonetheless, an SAR ADC traditionally requires good capacitor matching to achieve the resolution of 10 b or higher, which leads to large ADC size. This limits its application to the array sensor. The large capacitor array also leads to long settling, which limits the dynamic performance of the SAR ADC.

SAR ADC capacitor mismatch can be possibly compensated with on-chip calibration. Most SAR ADC calibration techniques perform the calibration in the analog domain, which requires extra analog circuits [26]–[29]. This extra analog complexity is unfavorable for the array implementation. In recent years, non-binary capacitor array has been developed to create redundancy so as to tolerate the decision error in high-speed SAR ADC [30], [31]. The first SAR ADC digital calibration scheme was based on perturbation over a nonbinary capacitor array [32].

In this work, we developed a novel SAR ADC digital calibration technique with the dithering method for high-performance ADC arrays. This technique measures weights of MSB (most significant bit) capacitors through randomized switching. With

Manuscript received November 09, 2011; revised April 21, 2012; accepted April 29, 2012. Date of publication June 25, 2012; date of current version August 21, 2012. This paper was approved by Associate Editor Boris Murmann. This work was supported in part by the ITC of Hong Kong Special Administrative Region Government under Grant TS/156/09 and by Huawei Technologies Ltd.

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Digital Object Identifier 10.1109/JSSC.2012.2198350

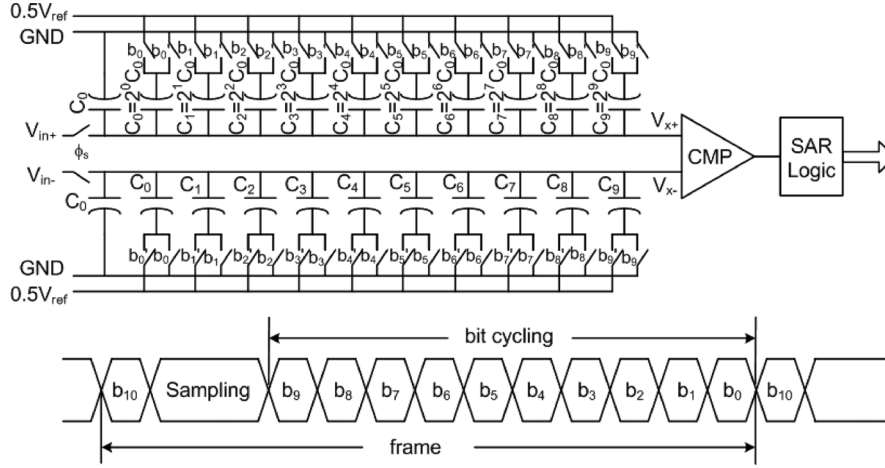


Fig. 1. 11-b top-plate-sampled binary-weighted SAR ADC.

accurately known capacitor weights, the capacitor mismatch in the array can be compensated. No extra analog calibration overhead is needed. As calibration control circuits are shared among all column SAR ADCs, on-chip calibration overhead is negligible. With much smaller capacitance, the dynamic performance of the ADC is also improved. The technique is implemented in a high-speed CIS, which includes an ADC array with 256 SAR ADCs.

This paper is organized in six sections. SAR ADC fundamentals are introduced in Section II. The dithering scheme is introduced in Section III. Detailed circuits are given in Section IV. Section V gives measurement results. Conclusions are drawn in Section VI.

II. CAPACITOR MISMATCH EFFECT IN SAR ADC

Unlike the conventional bottom-plate-sampled SAR ADC, top-plate-sampled binary-weighted SAR ADC as shown in Fig. 1 is often designed recently because it cuts the total capacitance by half [20], [25], [31]. The smaller input capacitance also enhances the dynamic performance of the ADC. As shown in Fig. 2, a bit cycling procedure keeps subtracting the binary weights from the input signal. At the end, the remaining voltage on V_x is given by

$$V_e = V_{in} - s_{10} \left(\sum_{i=0}^9 b_i W_i \right) \quad (1)$$

where s_{10} is the sign, which is either -1 or 1 , b_i is the quantized code from the SAR ADC for bit i , either 0 or 1 , and W_i is the capacitor weights. For top-plate-sampled binary-weighted SAR ADC with 11-b resolution, $W_i = (2^i/2^{10})V_{ref}$.

The capacitor array size can be reduced by including a bridge capacitor [27]–[29], [33], [34]. Fig. 3 shows the 11-b top-plate-sampled SAR ADC used in this work. The least significant bit (LSB) and most significant bit (MSB) sections are identical binary-weighted capacitor arrays. It operates similarly as the binary-weighted SAR ADC. Equation (1) is

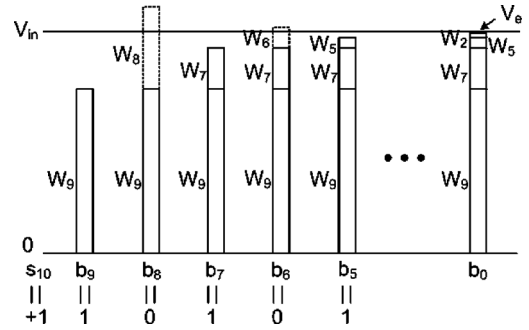


Fig. 2. Conventional bit cycling procedure ($b_{10} = 1, s_{10} = +1$).

still applicable, however, the weights for the MSB and LSB sections are given by

$$\begin{cases} W_{i,M} = \frac{C_i}{C_{MSB} + C_b || C_{LSB}} V_{ref} \\ = C_i \left(\frac{C_b + C_{LSB}}{\Lambda} \right) V_{ref}, & i = 5, 6, \dots, 9 \\ W_{i,L} = \frac{C_i}{C_{LSB} + C_b || C_{MSB}} \frac{C_b}{C_b + C_{MSB}} V_{ref} \\ = C_i \left(\frac{C_b}{\Lambda} \right) V_{ref}, & i = 0, 1, \dots, 4 \end{cases} \quad (2a)$$

$$\Lambda = C_{MSB}C_{LSB} + C_b(C_{MSB} + C_{LSB}) \quad (2b)$$

where C_{LSB} and C_{MSB} are the total capacitances in the LSB and MSB sections, respectively. If LSB and MSB sections have identical binary-weighted capacitors, and C_b matches the unitary capacitor C_0 , the weights are linear, which enables perfect linearity to this ADC.

The nonlinearity of the SAR ADC is caused by the non-linear weights due to the capacitor mismatch and parasitics. By considering only random mismatch of the capacitors, static linearity performances of the 11-b SAR ADC with binary-weighted capacitor array and bridged capacitor array in Figs. 1 and 3 are estimated, respectively, through Monte Carlo simulations for our $0.18\text{-}\mu\text{m}$ CMOS process. To achieve 0.5-LSB INL for 10-b resolution with 99% probability, the minimum-size metal–insulator–metal (MIM) capacitor, 20 fF ,

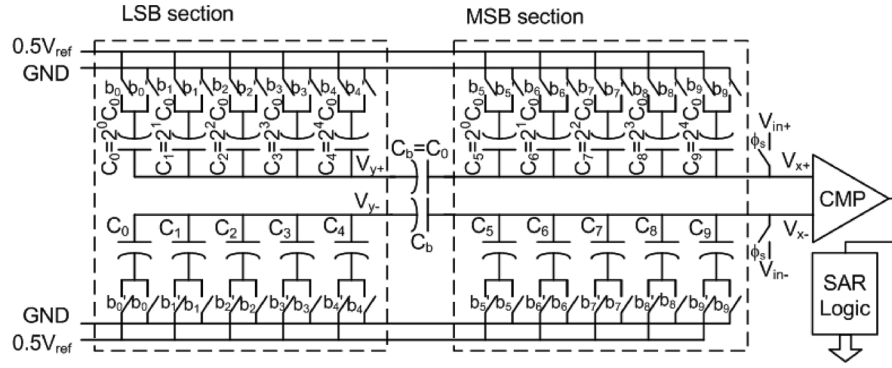


Fig. 3. 11-b top-plate-sampled SAR ADC with bridged capacitor array.

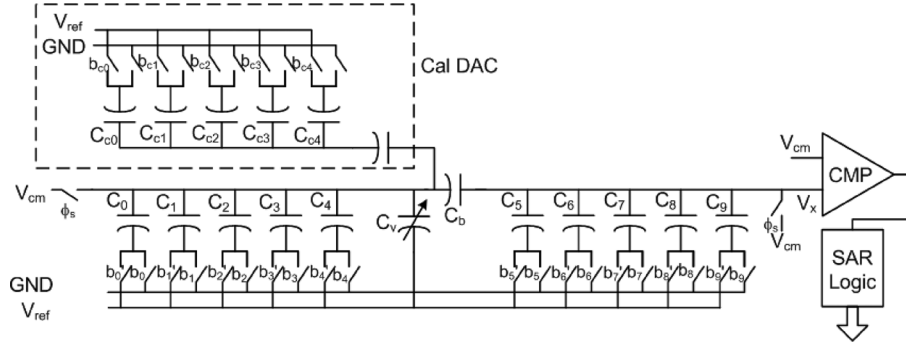


Fig. 4. SAR ADC calibration with a capacitive DAC: single-ended equivalent diagram.

can be used as the unitary capacitor C_0 for the binary-weighted capacitor array. For the bridged capacitor array, the unitary capacitor should be at least 80 fF due to the more stringent matching requirement. The total capacitor array sizes for the two arrays are $8.8 \times 10^4 \mu\text{m}^2$ and $1.4 \times 10^4 \mu\text{m}^2$, respectively. With metal–oxide–metal (MOM) capacitors, the absolute capacitance can be smaller. However, the physical area is similar due to the matching requirement. Apart from the capacitor mismatch, parasitics at V_y node will also significantly affect the linearity of the SAR ADC for bridged capacitor array [34]. Simulation results show that, for the bridged capacitor array in Fig. 3, the parasitic capacitance at V_y node should be smaller than 4.8% of the total LSB capacitance to achieve 10-b ENOB.

To further reduce the unitary capacitor, on-chip calibration methods have been applied to overcome the capacitor mismatch and parasitics. Analog calibration techniques compensate the capacitors in the array to their original binary weights by an extra digital-to-analog converter (DAC) with sub-LSB accuracy. The DAC can be either a voltage DAC [26] or a capacitive DAC [27]–[29]. In [28], to compensate for the mismatch in the SAR ADC with bridged capacitor array, the MSB capacitor mismatch is compensated by a calibration DAC, while C_b mismatch is compensated by a variable capacitor C_v , as shown in Fig. 4. Analog SAR ADC calibration techniques require extra analog circuits, which increases the circuit complexity [29].

Similar to pipelined ADCs, another direction to compensate for the capacitor mismatch is to measure the actual capacitor weights and to calculate the input voltage with the actual weights instead of the ideal binary weights. The recovery process can be performed in the digital domain. With calibra-

tion methods, the capacitor array size can be reduced towards the sampling noise limit.

III. SAR ADC CALIBRATION WITH DITHERING

Dithering was widely used in pipelined ADCs to measure errors from MDACs (multiplying DAC) in the background [35]–[37]. In [35], a pseudorandom noise (PN) signal is injected to the reference voltage path in the pipelined ADC. At the digital back-end, the known PN sequence is applied to correlate the pipelined ADC output. With sufficiently long PN sequence, the influence from the input signal of the pipelined ADC is removed after the correlation. Only the reference voltage error remains. Inspired by pipelined ADC dithering methods, in this work, we develop a dithering technique to measure the MSB capacitor weights for the SAR ADC. The dithering technique needs an efficient method to inject the dithering signal and to extract the capacitor weight in the dithered signal. Also, the reduced signal range due to dithering needs to be handled.

A. Dithering Method

In the SAR ADC in Fig. 3, the capacitor weight W_j can be conveniently added to the input signal by charging C_j through switch b' during the sampling phase and switching it to b during the bit cycling as shown in Fig. 5. By reversing the switching sequence, W_j is subtracted from the input signal. With a $\{1, -1\}$ PN sequence K , the dithered signal is quantized by the rest of the SAR ADC as shown by

$$V_{in} = KW_j = s_{10} \sum_{i=0, i \neq j}^9 b_i W_i + V_e. \quad (3)$$

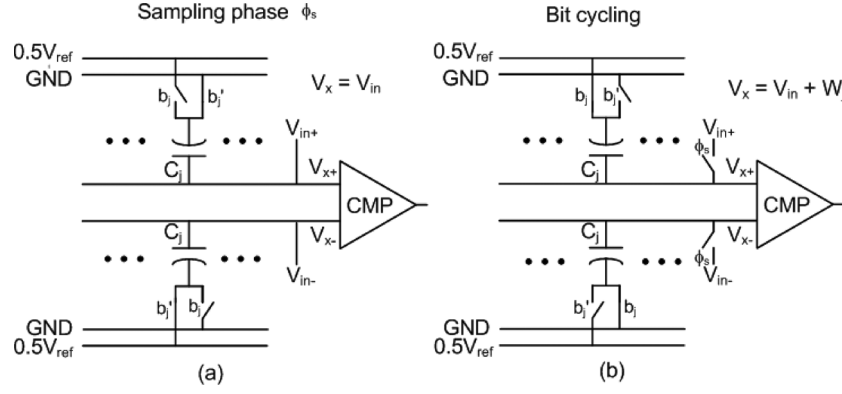


Fig. 5. Dithering signal injection. (a) Sampling phase. (b) Bit-cycling phase.

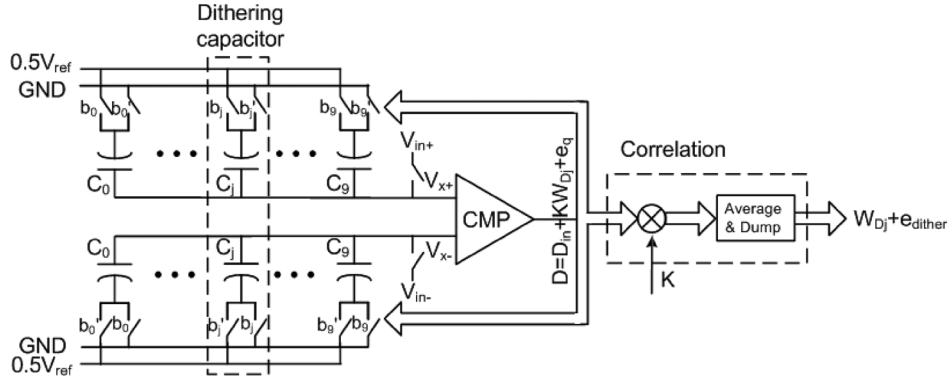


Fig. 6. Signal diagram of the dithering method for SAR ADC.

Correlating K with (3), the input voltage can be removed and W_j can be calculated in the digital domain.

To derive the digital calibration method, we can convert the analog weights (W_i) into digital weights (W_{Di}), which have the unit of codes, by dividing W_0 . Capacitors in the LSB section are regarded as ideal. Hence, their digital weights are regarded as $\{1, 2, 4, 8, 16\}$. For capacitors in the MSB section, W_{Di} includes both the mismatch of C_i and C_b , as well as the parasitics. Divided by W_0 , the quantization equation (3) for the dithered signal is then

$$\begin{aligned} D_{in} + KW_{Dj} &= s_{10} \left(\sum_{i=5, i \neq j}^9 b_i W_{Di} + \sum_{i=0}^4 b_i W_{Di} \right) + e_q \\ &= s_{10} \left(\sum_{i=5, i \neq j}^9 b_i W_{Di} + D_l \right) + e_q \end{aligned} \quad (4)$$

where D_l is the 5-b LSB code. e_q is the quantization error. As shown in Fig. 6, correlating the output code with the PN sequence K , (4) is converted into

$$\begin{aligned} \sum_{i=5, i \neq j}^9 \overline{K s_{10} b_i W_{Di}} + \overline{K s_{10} D_l} &= W_{Dj} + \overline{D_{in} K} - \overline{e_q K} \\ &= W_{Dj} + e_{dither}. \end{aligned} \quad (5)$$

By running this dithering procedure for all of the MSB capacitors, the following matrix equation is obtained:

$$\begin{bmatrix} 1 & -\overline{K_9 s_{10,9} b_{8,9}} & -\overline{K_9 s_{10,9} b_{7,9}} & \cdot & \cdot \\ -\overline{K_8 s_{10,8} b_{9,8}} & 1 & \cdot & \cdot & \cdot \\ -\overline{K_7 s_{10,7} b_{9,7}} & \cdot & 1 & \cdot & \cdot \\ \cdot & \cdot & \cdot & 1 & \cdot \\ \cdot & \cdot & \cdot & \cdot & 1 \end{bmatrix} \times \begin{bmatrix} W_{D9} \\ W_{D8} \\ W_{D7} \\ W_{D6} \\ W_{D5} \end{bmatrix} = \begin{bmatrix} \overline{K_9 s_{10,9} D_{l,9}} \\ \overline{K_8 s_{10,8} D_{l,8}} \\ \overline{K_7 s_{10,7} D_{l,7}} \\ \overline{K_6 s_{10,6} D_{l,6}} \\ \overline{K_5 s_{10,5} D_{l,5}} \end{bmatrix} - \begin{bmatrix} e_{dither9} \\ e_{dither8} \\ e_{dither7} \\ e_{dither6} \\ e_{dither5} \end{bmatrix}. \quad (6)$$

With a sufficiently long PN sequence, the dithering error e_{dither} reduces to zero. Then, $W_{D5} \sim W_{D9}$ can be calculated from (6) accurately. With MSB weights known, during the normal mode, the digital input voltage can be recovered with the exact weight for every MSB bit in real time as follows:

$$D_{in} = s_{10} \left(\sum_{i=5}^9 b_i W_{Di} + D_l \right). \quad (7)$$

B. Signal Range

One problem of the SAR dithering is that, if C_j is selected for dithering, it cannot be used for quantization in the bit cycling as noted in (4). Hence, the quantization error e_q at the end of the conventional bit cycling procedure can be larger than 1 1 LSB.

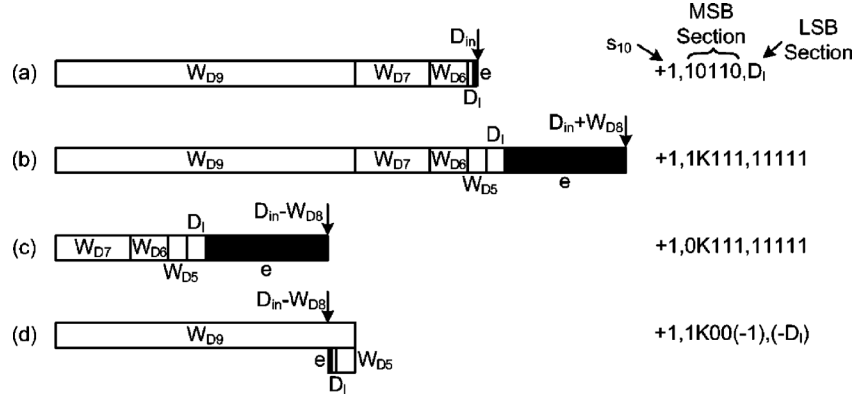


Fig. 7. (a)–(c) Code gap exists with the dithering scheme. (d) Modified bit cycling procedure to fill the code gap. ($s_{10} = +1$, C_8 is used for dithering).

If an input signal is quantized with $e_q > 1$ LSB, we call this input signal within a code gap. Fig. 7 illustrates the code gap if C_8 is selected for dithering. Fig. 7(a) shows the quantization of an input signal with digital code D_{in} . W_{D8} is not used in the quantization. If D_{in} is added with the dithering signal W_{D8} in Fig. 7(b), W_{D8} is needed following the conventional bit cycling. As W_{D8} is not available for quantization, the eventual quantized code is $+1, 1K111, 11111$ with a large quantization error. Fig. 7(c) shows D_{in} is subtracted with the dithering signal W_{D8} . W_{D8} is also needed following the conventional bit cycling. Without it, the quantized code is $+1, 0K111, 11111$ with a large error as well. Both scenarios in Fig. 7(b) and (c) cause large e_{dither} in (6) and reduce the estimation accuracy.

The code gap in Fig. 7(c) can be avoided by overcharging the capacitors in the sampling phase. Realizing that $W_{D8} < D_{in} - W_{D8} < W_{D9}$, $C_0 \sim C_7$ can be charged through switch b'_i during the sampling phase instead of b_i although s_{10} is $+1$. C_9 is still charged through switch b_9 . During the new bit cycling, W_{D9} is subtracted from the dithered signal, which turns V_x in Fig. 3 negative. Then, $W_{D0} \sim W_{D7}$ are added to the signal to quantize the negative difference $(D_{in} - W_{D8}) - W_{D9}$ as shown in the example in Fig. 7(d). As long as $\sum W_{Di(i=0:7)} > W_{D8}$, the error after this modified bit cycling procedure is within $[-LSB, 0]$. The eventual quantized code is $+1, 1K00(-1), (-D_l)$.

In general, with the MSB capacitor C_j used for dithering, 2^{9-j} gaps exist in the positive range. Fig. 8 shows the example when C_6 is used for dithering. When the input signal is dithered by $+W_{D6}$ or $-W_{D6}$, the dithered signal can possibly fall into the gray code gaps following the original bit cycling procedure. In these gray code gaps, lower-weighted bits are all 1. Except for the highest gap, all code gaps in the gray region can be avoided by overcharging the higher weighted capacitors $C_{j(i=j+1:9)}$ during the sampling phase. The negative voltage D_{res} can be quantized by the negatively charged lower weighted capacitors $C_{j(i=0:j-1)}$. As long as the sum of lower weights is larger than its own weight, i.e., $\sum W_{Di(i=0:j-1)} > W_{Dj}$, the quantization error e_q is always within $[-LSB, 0]$. To enable this condition, the bridge capacitor C_b is set to be a little bit larger than the unitary capacitor C_0 in our design. This can be easily derived from (2). Similar techniques have been used in analog calibration methods for SAR ADCs [27]–[29].

As shown in Fig. 8, the highest code gap cannot be avoided as it is out of the range, which is also the case in Fig. 7(b). Overcharging is not possible. As the input voltage can possibly be dithered with either $+W_{Dj}$ or $-W_{Dj}$, the maximum allowed input voltage to the SAR ADC for dithering is the sum of the weights of all the capacitors except for the dithered capacitor ($\sum W_{Di(i=j+1:9)} + \sum W_{Di(i=0:j-1)} - W_j$). As long as the input voltage is less than this value, the dithered signal can always be quantized with error within $[-LSB, LSB]$. Similar range can be derived for negative signals.

C. Dithering Procedure

If C_j is selected for dithering, it is charged through switch b'/b when K is $1/-1$ in the sampling phase and is switched to b/b' after the sampling phase so that the dithered signal to the SAR ADC is $V_{in} + W_j/V_{in} - W_j$. The dithered signal will be quantized twice. During the first quantization, capacitors are charged according to s_{10} in the sampling phase and follow the conventional bit cycling procedure afterwards to quantize the dithered signal. During the second quantization, higher-weighted capacitors ($C_{j+1} \sim C_9$) are charged according to s_{10} in the sampling phase, while lower-weighted capacitors ($C_0 \sim C_{j-1}$) are charged to the opposite polarity. After the sampling phase, the SAR ADC follows the modified bit cycling procedure in Fig. 7(d) to quantize the dithered signal. The first quantized code will be adopted unless it is within a code gap. In this case, the second quantized code will be adopted. The code will be used in (6). This process will be performed with a sequence of K for every MSB capacitor.

This calibration method can run in the background with an additional SHA. In this case, the quantization rate of the SAR ADC should double the Nyquist frequency to enable the double quantization. In this work, the SHA is not implemented on chip for simplicity. The input signal is generated with off-chip DAC and holds for two ADC cycles.

D. Correlation Length

The accuracy of the dithering technique is determined by the correlation between the quantization error (e_q) and the PN sequence (K) in (5). The dithering error e_{dither} in (5) is mainly caused by this correlation. This correlation is studied

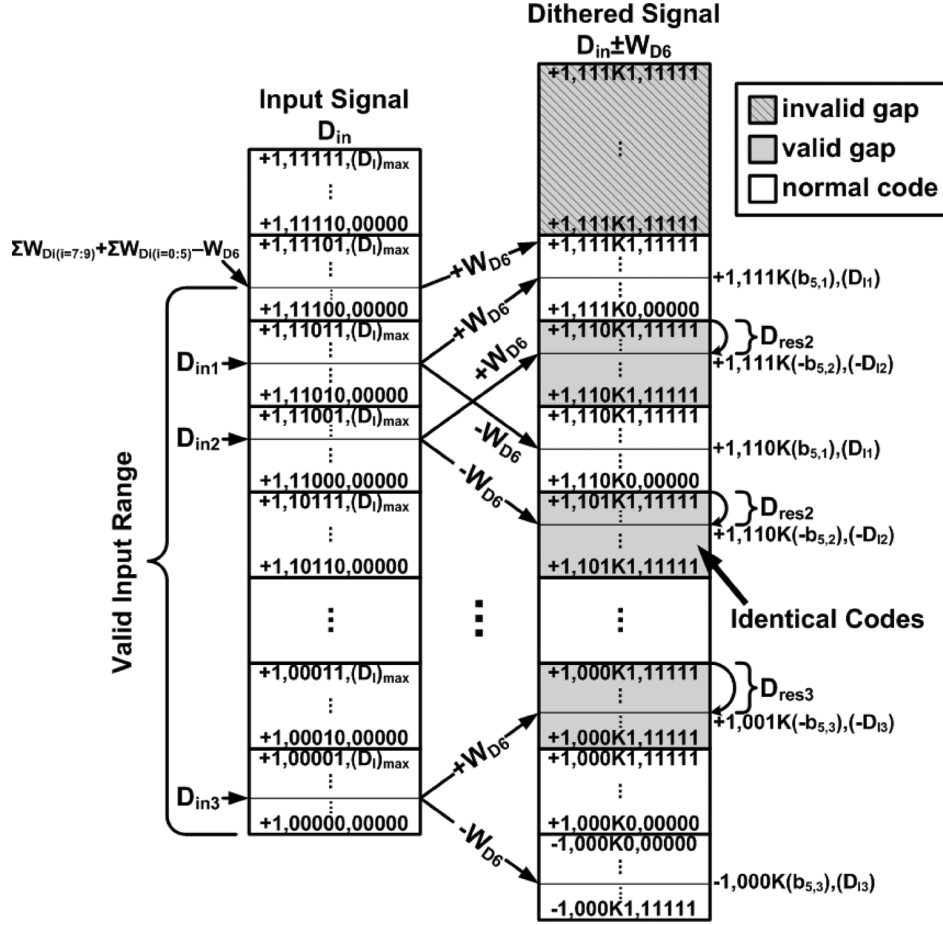
Fig. 8. Code gap existence when C_6 is used for dithering.

TABLE I
THE MSB CAPACITORS WEIGHTS IN SIMULATIONS AND THE MEASURED VALUES AFTER DITHERING

	W_{D5}	W_{D6}	W_{D7}	W_{D8}	W_{D9}
Set	23.91	46.48	88.55	172.72	393.18
Calculated	23.92	46.45	89.64	174.04	395.50

TABLE II
THE DITHERING ERROR AND THE ESTIMATED CAPACITOR WEIGHT ERROR

Number	2^{12}	2^{14}	2^{16}	2^{18}
$e_{dither9}$	0.0014	0.0015	0.0014	0.0014
e_{w9}	42.6%	4.0%	1.0%	0.6%

with MATLAB simulations. In the SAR ADC, MSB capacitors and C_b are added with random mismatches. One set of MSB capacitor weights is listed in Table I. In simulations, the dithering procedure is performed on this SAR ADC. When C_9 is selected for dithering, the simulated dithering error ($e_{dither9}$) with different correlation lengths is listed in Table II. With more than 2^{12} samples, the dithering error is normally less than 0.002. e_{w9} is the error of W_{D9} calculated with (6). When the PN sequence is short, $\overline{KsD_l}$ in (6) has large variation. For some sequence, this term can be small so that even a small e_{dither} can lead to large estimation error, which is the case for the 42.6% error in Table II. With longer sequence, the $\overline{KsD_l}$ term is larger and more consistent, which reduces the effect of e_{dither} .

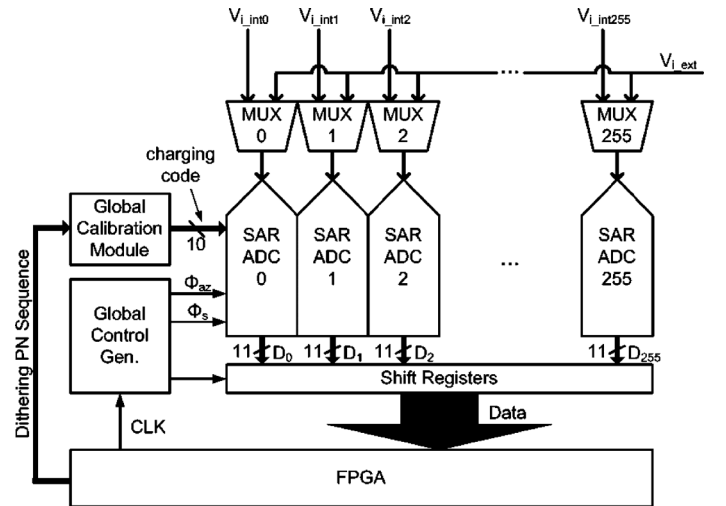


Fig. 9. Architecture diagram of the SAR ADC array.

Without knowing the accurate MSB weights, the DNL and INL of the SAR ADC with the MSB weights listed in Table I are 3.46 LSB and 41.78 LSB, respectively. The ENOB for a signal at the Nyquist frequency is 5.6 bits. With 2^{18} dithering samples, the calculated MSB weights from (6) are within 0.6% of the actual values. The calculated $W_{D5} \sim W_{D9}$ with 2^{18} dithering samples are shown in Table I. With these measured weights, the

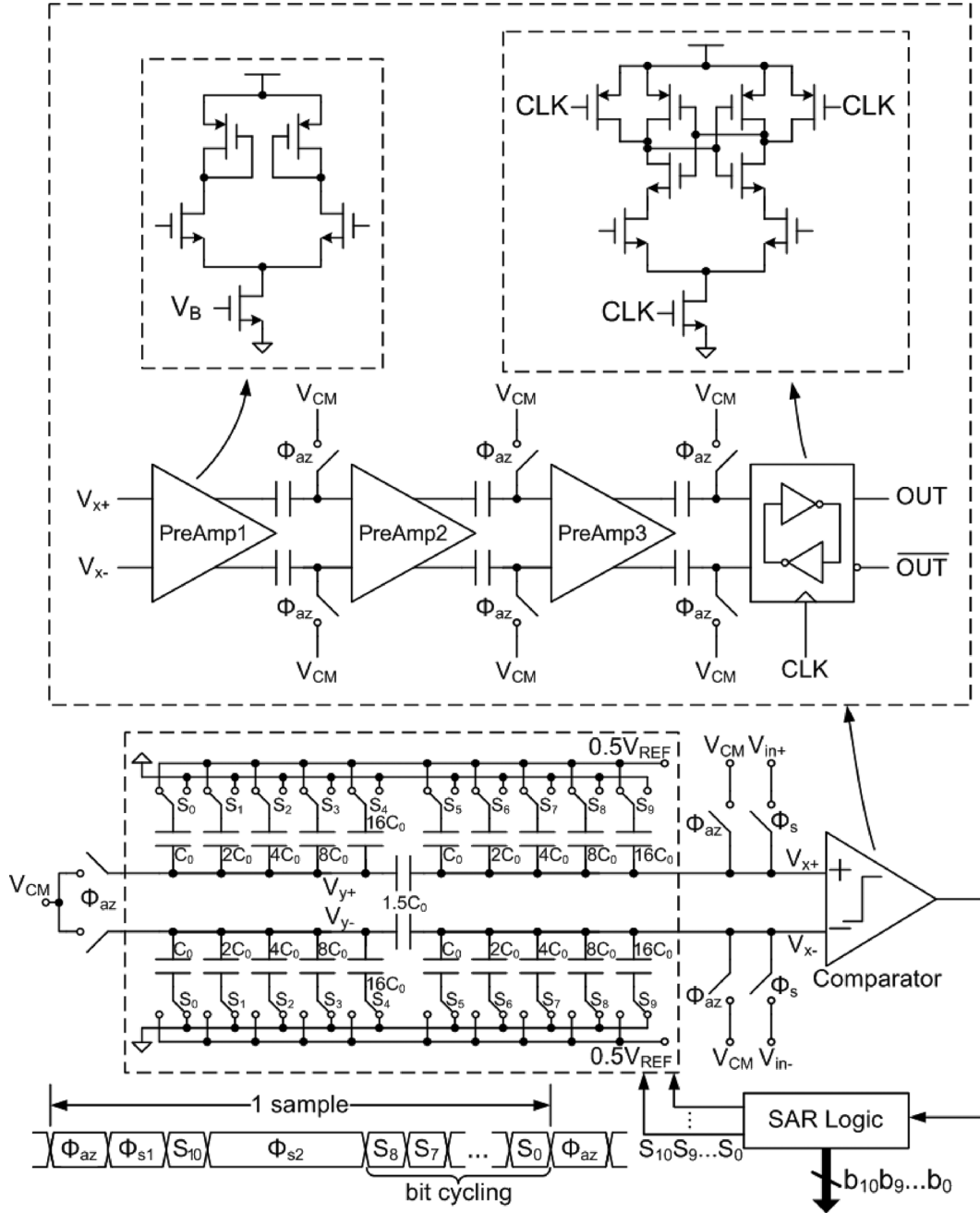


Fig. 10. Circuit diagram of the SAR ADC.

DNL and INL of the SAR ADC are 1.0 LSB and 1.0 LSB. The ENOB for the same signal is improved to 9.8 b.

E. Digital Decoders

This calibration procedure requires two decoders. The first decoder implements (6) in the calibration mode. As shown in (6), the correlation terms can separate into single-bit correlations in the MSB matrix and 5-b correlations of the LSB codes. These calculations need single-bit accumulators and 5-b adders with carry-in. The 5×5 matrix inverse and the floating-point multiplications/additions followed seem to be computationally expensive. However, as only one such calculation is required over the whole calibration procedure, the hardware overhead is

negligible as all multiplications and additions can be performed sequentially.

The real-time addition in (7) dominates the hardware overhead of this calibration method. This addition can be done with a 14-b fixed-point (11-bit integer, 3-b fractional number) adder running at the speed of $6f_{\text{ADC}}$. Besides, it requires 70-b SRAM to store the 14-b $W_{D5} \sim W_{D9}$. After synthesis, the two decoders need 2233 logic gates. In 65-nm CMOS processes, which are commonly used for digital signal processors, this needs $2363\text{-}\mu\text{m}^2$ die area with $1.4\text{-}\mu\text{W}$ power for our 768-kS/s quantization rate.

The calibration procedure in Section III-C also needs extra decoder to generate the control signals for the sampling and

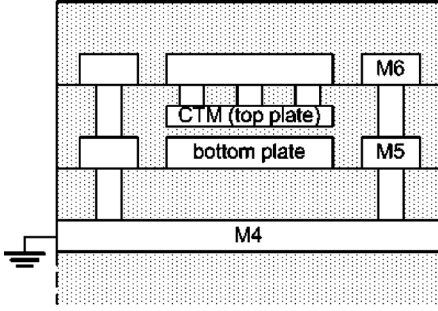


Fig. 11. Cross section of a unitary capacitor.

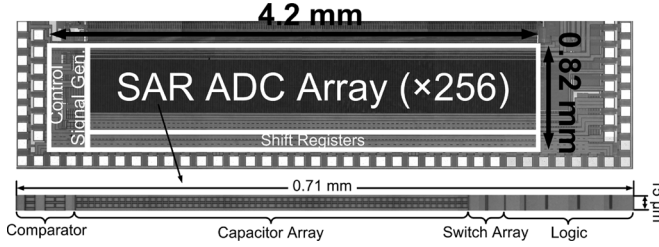


Fig. 12. Micrograph of the SAR ADC array chip.

bit-cycling phases in the calibration mode. As these control signals can be shared among different channels, the average cost is negligible.

IV. CIRCUIT DESIGN

The SAR ADC array is designed in a $0.18\text{-}\mu\text{m}$ CMOS process for a high-speed CMOS imaging sensor. The 256×256 CIS requires 256 SAR ADCs to quantize at 768 kS/s with 10-b resolution. The signal range is $\pm 2\text{ V}$ (differential). The challenge is to design this ADC within the $15\text{-}\mu\text{m}$ column pitch with good reliability. The architecture of the ADC array is shown in Fig. 9. The ADC channel quantizes the CIS column data. During the testing mode, the input signal is supplied externally. Outputs from the ADC array are captured and processed with an off-chip digital processor. The digital processor also generates the PN sequence. The global calibration controller sets the charging codes of the ADCs in the sampling phase.

Detailed circuits of one SAR ADC are shown in Fig. 10. The capacitor array is designed with a bridge capacitor. Due to area constraints, MIM capacitors are used for the array. Simulations in Section II show that 80-fF unitary capacitor is needed for 10-b linearity for this bridged capacitor array, which leads to 10-pF overall capacitance. With calibration, the minimum MIM capacitor (20 fF) in the process is used for the unitary capacitor. The total capacitance is only 2.54 pF. The 20-fF MIM capacitor has about 1% mismatch. Simulations show that the linearity of this SAR ADC is about 8.3-b ENOB with this capacitor mismatch. C_b is designed to be $1.5C_0$ to overdesign weights of LSB capacitors. To achieve good supply rejection in the large ADC array, the ADC is designed with fully differential circuits. Only one reference voltage V_{ref} is needed for the input signal range. V_{cm} is used to set the input common-mode voltage of the preamplifiers in the comparator. It is also used to reset the floating node V_y during the sampling phase to avoid charge accumulation. Due to the fully-differential design of the capacitor array,

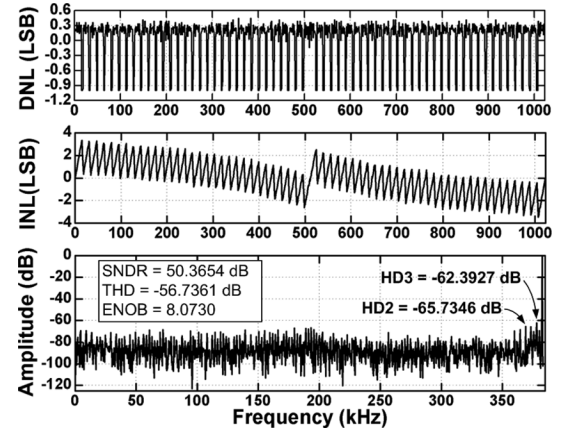


Fig. 13. Measured performance of an SAR ADC without calibration. (a) DNL. (b) INL. (c) Spectrum of a 380-kHz sinusoidal signal.

TABLE III
MEASURED WEIGHTS OF MSB CAPACITORS IN CHANNEL 140

	W_{D5}	W_{D6}	W_{D7}	W_{D8}	W_{D9}
Measured	26.35	52.90	105.85	211.55	422.65

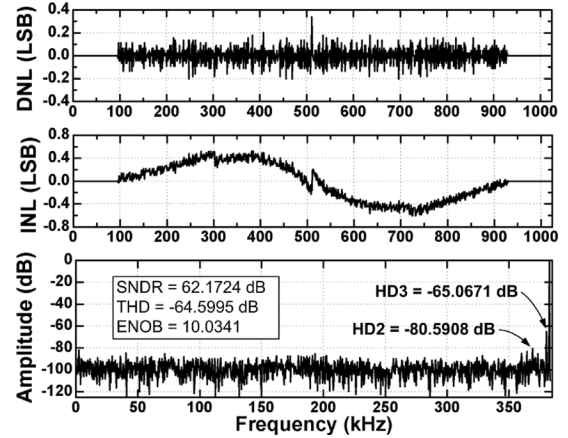


Fig. 14. Measured performance of an SAR ADC with calibration. (a) DNL. (b) INL. (c) Spectrum of a 380-kHz full-scale sinusoidal signal.

the common-mode signal to the comparator remains constant at V_{cm} during the bit cycling.

To further minimize the area, no matching is applied in the capacitor array layout. Nonetheless, to avoid cross-couplings between the bottom plates and top plates of capacitors in different sections, shielding is used. The cross section of a capacitor is illustrated in Fig. 11. The capacitor is encapsulated in a grounded well formed by M4, M5 and M6.

The comparator includes three preamplifiers with auto-zeroing. The gain of the preamplifier is 10 dB. The comparator offset is suppressed to be lower than 1 mV. Due to the constant common mode input to the comparator, the comparator offset is mostly signal-independent. The comparator uses 3.3-V supply.

To reduce the power consumption, the logic circuits are designed with 1.8-V supply. The SAR controller implements the modified bit cycling. During the calibration mode, before the sampling phase, a 10-b charging code is written into a register in the logic circuits, which determines the capacitor charging

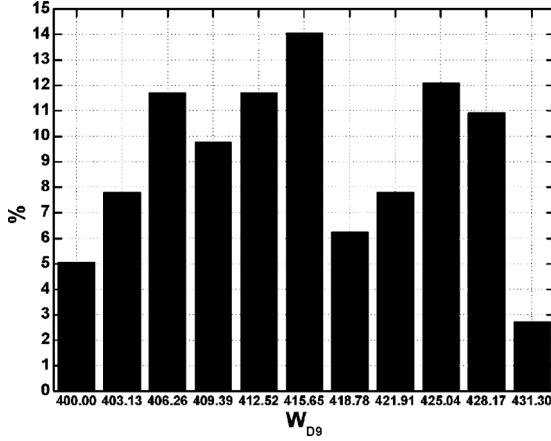
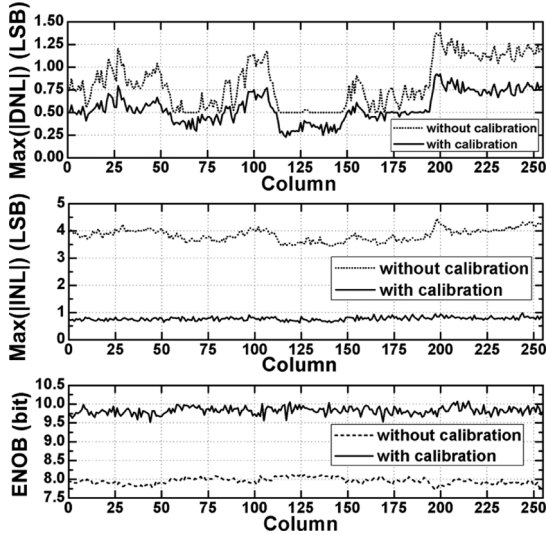
Fig. 15. Distribution of the digital weight of capacitors C_9 in the array.

Fig. 16. Measured performance of the SAR ADC array with and without calibration. (a) Maximum DNL. (b) Maximum INL. (c) ENOB with a 380-kHz sinusoidal signal.

pattern in the sampling phase. This charging code includes the dithering bit. It is generated by the global calibration module.

V. MEASUREMENTS

The micrograph of fabricated SAR ADC array is shown in Fig. 12. The active area of the ADC array as well as the global controller and shift registers cover $4.2 \text{ mm} \times 0.82 \text{ mm}$. A single SAR ADC occupies $15 \mu\text{m} \times 710 \mu\text{m} \approx 0.01 \text{ mm}^2$. Within this size, the capacitor array occupies 62.3% area ($6.6 \times 10^3 \mu\text{m}^2$). The comparator occupies 9.3% area. The switch array occupies 5.7% area. The logic circuits occupy 22.7% area.

To measure the ADC array, the external analog signal is selected. This signal is generated by a 20-b DAC on the testing board. The array output is captured by a Xilinx FPGA board. The FPGA board generates the dithering PN sequence, and sends it to the global calibration module. The calibration decoders are implemented off-chip in software. Measurements of the imaging chip can be found in [38], which specifically introduces the high-speed CIS design.

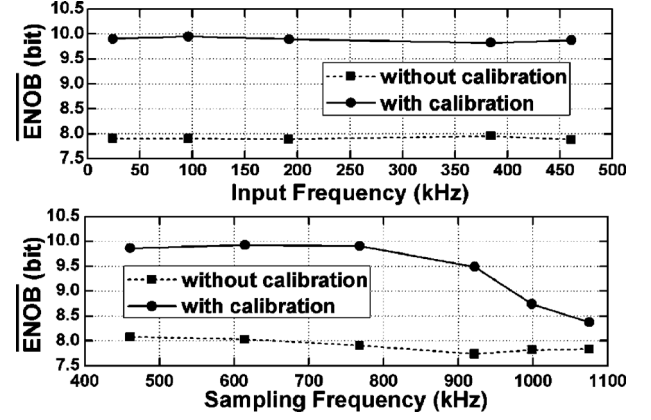
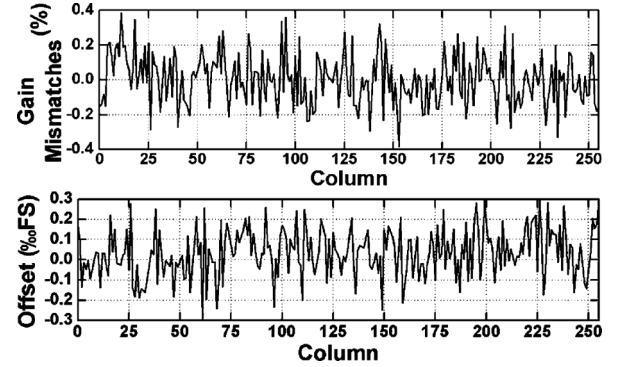
Fig. 17. $\overline{\text{ENOB}}$ among all of the columns: (a) versus input frequency at 768 kS/s and (b) versus sampling frequency with 24-kHz input sinusoid.

Fig. 18. Column mismatches after the SAR ADC calibration: (a) gain mismatches relative to the mean value and (b) column-input-referred offset relative to the full signal range.

TABLE IV
POWER CONSUMPTION OF SAR ADC COMPONENTS (μW)

Preamplifier	18.0
Latch	2.3
V_{ref}	3.0
V_{bias} Gen. (/256)	0.1
Local control	34.2
Global control (/256)	0.4
Total	58.0

TABLE V
PERFORMANCE SUMMARY OF THE ADC ARRAY

Process	0.18- μm CMOS Process	
Sampling Frequency	768 kS/s	
Supply Voltage	3.3 V (analog), 1.8 V (digital)	
ADC Array Size	256 channels	
Single ADC Size	$15 \mu\text{m} \times 710 \mu\text{m} \approx 0.01 \text{ mm}^2$	
Single ADC Power	58.0 μW	
Single ADC FOM	74 fJ/conversion-step	
	Before Cal.	After Cal.
DNL (LSB)	0.82	0.55
INL (LSB)	3.85	0.77
SNDR (dB)	49.7	60.9
ENOB (bit)	7.96	9.83

The histogram method is used to measure a typical SAR ADC channel (Channel 140). The measured DNL and INL are shown in Fig. 13(a) and (b). The maximum DNL and INL

TABLE VI
PERFORMANCE COMPARISONS WITH BENCHMARK SAR ADC DESIGNS

	Verma, JSSC 2007, [23]	Agnes, ISSCC 2008, [33]	Liu, ISSCC 2010, [32]	Kang, VLSI 2009, [40]	Liu, VLSI 2010, [41]	Walker, VLSI 2011, [2]	Yoshioka, ISSCC 2010, [28]	Liu, ISSCC 2010, [31]	Liu, JSSC 2010, [25]	Zhu, JSSC 2010, [34]	This work
Process	0.18- μ m CMOS	0.18- μ m CMOS	0.13- μ m CMOS	0.13- μ m CMOS	0.18- μ m CMOS	0.13- μ m CMOS	65-nm CMOS	65-nm CMOS	0.13- μ m CMOS	90-nm CMOS	0.18- μ m CMOS
ENOB (bit)	10.5	9.3	10.9	10.2	9.7	9.7	9.1	9.0	9.2	9.1	10.0
Sampling Frequency (kS/s)	100	100	45,000	11,000	10,000	31	50,000	100,000	50,000	100,000	768
Power (μ W)	25	3.6	3,020	3,570	98	1.1	820	1,130	826	3,000	58
FOM (fJ/step)	172	55	36	281	11	41	30	22	28	55	74
Area (mm^2)	0.65	0.24	0.06	0.7	0.09	0.11	0.04	0.026	0.05	0.18	0.01
Area Efficiency ($\mu\text{m}^2/\text{code}$)	449	381	31	595	108	132	73	50.8	85	328	9.8

TABLE VII
PERFORMANCE COMPARISONS WITH OTHER ADC ARRAY DESIGNS

	Walker, VLSI 2011, [2]	Snoei, JSSC 2006, [4]	Yuan, JSSC 2009, [6]	Krymski, VLSI 1999, [8]	Furuta, JSSC 2007, [10]	Nitta, ISSCC 2006, [7]	This Work
Application	Neural Recording	Imager	Imager	Imager	Imager	Imager	Imager
Type	SAR	Single Slope	Single Slope	SAR	Cyclic	Single Slope	SAR
Process	0.13- μ m CMOS	0.18- μ m CMOS	0.35- μ m CMOS	0.5- μ m CMOS	0.25- μ m CMOS	0.18- μ m CMOS	0.18- μ m CMOS
ENOB (bit)	9.72	10.0	8.2	7	12	12b@65KS/s 10b@260KS/s	10.0
Sampling Frequency (kS/s)	31.3	NA	0.48	500	2000		768
Power (μ W)	1.1	NA	NA	NA	430	NA	58
FOM (fJ/step)	41.5	NA	NA	NA	52.5	NA	74
Area (μm^2)	110,000	7,900	10,500	39,000	88,000	5,700	10,600
Area Efficiency ($\mu\text{m}^2/\text{code}$)	130.4	7.8	35.9	304.7	21.5	5.6	9.8
# of Channels	96	340	32	1024	512	1920	256

are 1.0 LSB and 3.5 LSB respectively. Both DNL and INL show significant code gaps, which indicate the uneven MSB capacitors. A 380-kHz sinusoidal full-scale signal is applied to the ADC. The digitized spectrum is shown in Fig. 13(c). The ADC has high tones. The THD (total harmonic distortion) is -56.7 dB. The signal-to-noise-and-distortion ratio (SNDR) is 50.4 dB. The overall ENOB is 8.1 b.

The dithering method is run to measure the MSB capacitors. After collecting one million dithering samples, (6) is followed for the calculation. The calculated weights of MSB capacitors in Channel 140 are listed in Table III. With these measured weights, the DNL and INL in Fig. 13 are compensated with (7). The calibrated DNL and INL are shown in Fig. 14(a) and (b). As weights of MSB capacitors are smaller than their binary values, the code range is about 18.3% smaller. The DNL and INL are significantly suppressed to 0.34 LSB and 0.60 LSB. Similarly, the digital spectrum is improved. Most tones are suppressed. HD3 dominates. The THD is improved to -64.6 dB. The SNDR is improved to 62.2 dB. The ENOB is 10.0 b.

With this calibration method, MSB capacitor weights of all channels are measured in parallel. As an example, the weight distribution of capacitors C_9 in the array is shown in Fig. 15.

From measurements of the 256 channels, the mismatch of W_{D9} in the array is about 7.5%. As W_{D9} is calculated from its lower weights, this number includes accumulated contributions from all MSB capacitor mismatch, all LSB capacitor mismatch, C_b mismatch, and parasitic capacitance mismatch among different channels. The performance of all channels in the array is shown in Fig. 16. Without calibration, all channels have ENOB lower than 8.1 bits. After calibration, all channels have ENOB better than 9.5 b. The average ENOB among all columns ($\overline{\text{ENOB}}$) is measured with different input signal frequencies and sampling frequencies in Fig. 17. In measurements in Fig. 17(a), the sampling rate is fixed at 768 kS/s. In measurements in Fig. 17(b), the input signal frequency is fixed at 24 kHz. After calibration, $\overline{\text{ENOB}}$ maintains around 9.8 b up to the Nyquist frequency. $\overline{\text{ENOB}}$ begins to drop when the sampling frequency is higher than 800 kHz. In the CIS, an adaptive column-amplifier front-end is added in front of the SAR ADC array to increase the dynamic range [38]. After calibration of the SAR ADC array, the column gain variation and input-referred offset for the entire column are measured as shown in Fig. 18. The calibration can remove mismatch among SAR ADCs as the same external signal is used for all columns during the

calibration. Hence, these measured mismatches are mainly from column-amplifiers. The column-to-column fixed-pattern noise (FPN) due to the gain and offset variations is around 1% (peak-to-peak). With gain and offset known for every column, the FPN can be further compensated for imaging.

One SAR ADC channel consumes 58- μ W power. The partition of the power consumption is listed in Table IV. Power consumed by calibration decoders and the additional SHA is not included. Most power is consumed by logic circuits, which consumes 59.4% of the power. The figure-of-merit (FOM) of the SAR ADC is 74 fJ/conversion $\cdot$ step. The FOM can be improved by designing more power-efficient logic circuits. In the current design, static logic circuits are used. Performance of the fabricated ADC array is summarized in Table V. With the dithering technique, the good linearity and dynamic performance can be achieved reliably across all channels. Among the six CIS chips we measured, all channels can achieve ENOB better than 9.5 b after calibration.

Performance of the SAR ADC in this work is compared with benchmark SAR ADC designs reported in JSSC, ISSCC, and VLSI [39] in Table VI. Table VI includes all the SAR ADC designs with ENOB higher than 9 b. The power efficiency of this work is not among the best, which is mainly due to the high supply voltage. But this design has outstanding area efficiency. It has the smallest size for 10-b SAR ADC. It is 2.6 times smaller than the smallest design in the past [31]. As the capacitor array area is normally proportional only to the capacitor array resolution for SAR ADC, we define the area efficiency as

$$AE = \frac{\text{area}}{2^{\text{ENOB}}} \quad (8)$$

This efficiency is important for ADC arrays. Our design has 3.2 times better area efficiency than the most area efficient design [32] in the past. This is mainly due to the much smaller capacitor array.

The SAR ADC in this work is also compared with previous column ADC designs for array applications in Table VII. Compared with previous ADC array designs with SAR ADC [2], [8], our design has higher sampling rate and smaller area. Our SAR ADC has better efficiencies in both power and area, compared to the cyclic ADC [10]. The area efficiency of this design is comparable to the single-slope designs [4], [6], [7] while it achieves higher sampling rate. This design can also be easily integrated with high-speed digital processing circuits due to much simplified analog circuits.

VI. CONCLUSION

This paper reports a novel digital calibration technique for SAR ADCs. By randomly switching the capacitor array, dithering signals can be injected into the input signal. By arranging the capacitor charging and switching sequence, the input signal range can be expanded to avoid code gaps. With dithering, weights of MSB capacitors can be known accurately in the digital domain. This calibration method does not require any analog calibration circuit. The digital decoders are simple. With this calibration technique, very small capacitor array can be used in SAR ADCs without the need of matching. This can

lead to better area and power efficiencies, which is desired for array applications.

The new SAR ADC design is implemented in an ADC array including 256 SAR ADCs for a high-speed CMOS imaging sensor in a 0.18- μ m CMOS process. The size of a single ADC is only 710 μ m $\times$ 15 μ m, which is the minimum among SAR ADCs with about 10-b ENOB. This ADC array samples at 768 kS/s. Original ADCs have 0.82-LSB peak DNL and 3.85-LSB peak INL averaged across the array. For the 380-kHz full-scale sinusoidal signal, original ADCs have 7.96-b average ENOB. After dithering, peak DNL and peak INL of the ADCs averaged across the array are improved to 0.55 LSB and 0.77 LSB respectively. The average ENOB for the same Nyquist-frequency signal is 9.83 bits. The FOM of the SAR ADC is 74 fJ/conversion $\cdot$ step.

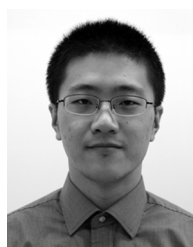
ACKNOWLEDGMENT

The authors would like to thank S. F. Luk, Hong Kong University of Science and Technology, for the chip bonding and testing.

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